Term Project Option: Finding the position of ancient North America and the age of dikes in the Middlesex Fells
EPS/ESE 102

**Possible science questions:**

1. What is the angle (in degrees from horizontal) of the magnetic field recorded by the dikes in the Middlesex Fells?
2. What is the paleolatitude of North America corresponding to this angle?
3. What is the age of the dikes (i.e. how long ago did they intrude into the bedrock)?
4. What was the polarity of the Earth's geodynamo at this ancient time?

**Basic steps:**

1. Before field trip, practice with phone guide, measuring magnetic fields at a consistent speed.
2. At Middlesex Fells, first use your phone hand-held to get a sense of the magnetic profile over the dike and contact regions. Take note of where the largest anomalies are.
3. Establish background levels of noise using the meter sticks, tape, and phone guide, i.e., the same tools you will use for the actual data profiles.
4. Take quantitative transects over the anomalies. Take as many datasets as possible.
5. Visualize the data as magnetic field intensity and direction maps in Python.
6. Address scientific questions with a variety of regression and hypothesis testing tools. **Focus will likely be on parameter estimation, non-linear regression, and Monte Carlo methods for error estimation.**
7. Visualization, analysis, and code will be submitted in a 3-page mid-term report and a 6-page final report. Results will be shared with class in a 6-minute mid-term presentation and a 10-minute final presentation.

**Key dates:**

1. September 9$^{th}$: Time allocated in class to practice with measurements and to discuss with your group.
2. September 19$^{th}$: Fieldtrip to Middlesex Fells. Meet at 8:45 am in the circle in front of Geological Museum (20 Oxford St). Leaving at 9:00 am sharp! Expect to be back by 2:00 pm. Lunch will be covered. Please bring comfortable clothes, shoes, notebook and pen, water.
3. October 19$^{th}$: "Cross pollination day". Class time will be allocated groups to talk to each other (especially the other group(s) doing this project) and teaching staff about direction of work.
4. Between October 19$^{th}$ and 21$^{st}$ (the 1.5 weeks before mid-term presentations): Individualized meetings of each group with teaching staff about progress and presentation.
5. October 28$^{th}$: 6-minute presentation to the class focusing on (1) the goals of the project, (2) the type of observations, (3) your observation strategy, (4) data analysis with the techniques covered so far in class, and (5) open questions and ideas about how to answer them. Mid-term report due at 5 pm November 2$^{nd}$ (after next class).

6. During Reading Period, date TBD: 10-minute final presentation to class. Should include more analysis and a more finalized set of results and interpretation. Final report due 5 pm 12/9/2026 (last day of Reading Period)
7. Expect to schedule a 20 minute meeting with teaching staff between Final presentation and report deadline to discuss the project feedback. This meeting is also a graded component of the project to ensure that you understand the underlying statistical principles even if you used AI during the project.

**Introduction:**

The surface of the Earth has been partitioned into distinct, mutually moving plates since at last 3.3 billion years ago (out of a total age of 4.5 billion years). Typically plates move at rates between 1 and 10 cm per year, or 0.1-1 degree of arc over the Earth's surface per million years. The magnetic field of the Earth, as any user of a magnetic compass knows, points approximately towards the poles. A lesser-known fact is that the *dip*, or angle to the horizontal, of the field changes systematically between the equator to the pole.

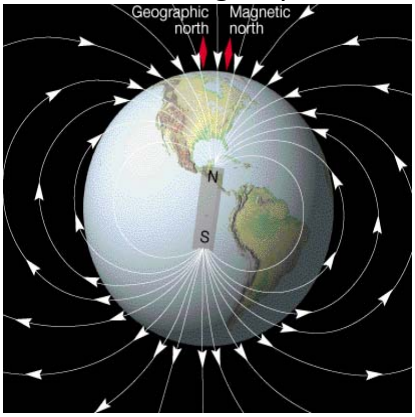



Figure 1: Schematic showing the geometry of the Earth's magnetic field. Image credit: thisoldearth.net

As seen in Fig. 1, the magnetic field is more parallel to the surface (i.e., smaller dip) at the pole and more vertical towards the poles. The relationship between the dip of the local field and the local latitude is:

$$2\tan(\lambda) = \tan(\alpha)$$

Where $\lambda$ is the latitude and $\alpha$ is the dip (again, angle between the horizontal plane and the local magnetic field). If we can measure $\alpha$ for a certain point in time in the distant past, we can compute the ancient latitude of that particular locality.

This basic principle has been used to reconstruct the motion of continents all the way back to, in some cases, more than two billion years in the past. Relevant for us is the motion of North America, or at least most of what today we call North America, and a "ribbon continent" called Avalonia.

Why is Avalonia important? You're standing on it right now!

As you can see in Fig. 2, Avalonia split off ("rifted" in the jargon) from West Africa around 500 million years ago (Ma) and drifted from about 60˚S to 20˚S latitude to dock with the east coast of North America at around 430 Ma.

While Avalonia was undergoing this drift, it is believed that dark, highly magnetic igneous rocks, known as basalt, intruded the area now belonging to the Middlesex Fells. As seen in the geologic map in Fig. 3, these dikes (colored green) trend roughly east to west and intrude a much older, less magnetic rock unit called the Lynn Volcanic Complex.

Rocks, including these dikes in Fig. 3, record the direction of magnetic field as a magnetization when they cool from a magma. This is a good opportunity to make an important distinction between the concepts of *magnetic field* and *magnetization*. Magnetic fields are what you detect with your phone or a magnetic compass. They are generated by electric currents or electron spins and propagate through free space, decreasing in intensity away from the source. Magnetization is a material property- that is, solid materials like rocks have a magnetization that, fundamentally, arises from the alignment of electrons within them. In this project (and almost any application in geomagnetism), **you are measuring magnetic field, but the quantity of interest is magnetization.** You are using statistical inference to quantify the latter.

All of this means that we have an interesting opportunity with these dikes: **If we can use magnetic field measurements to quantify their magnetization, specifically the dip of their direction, we can back out the latitude at which they erupted and, therefore, the time of their eruption following the trajectory in Fig. 2.**

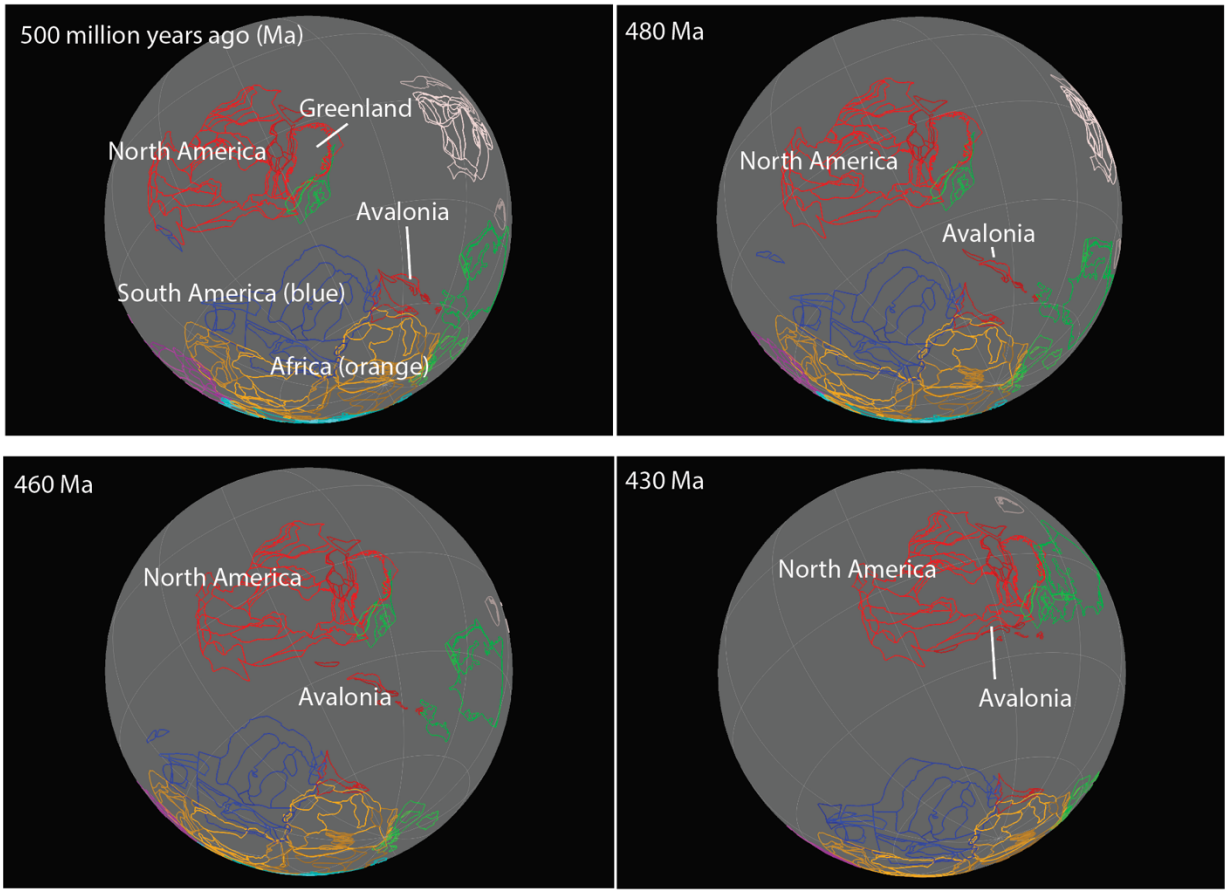



Figure 2: Reconstruction of continents on Earth between 500 Ma and 430 Ma. Reconstruction credit Merdith et al. (2021) *Earth Sci. Rev.* 214, 103477.

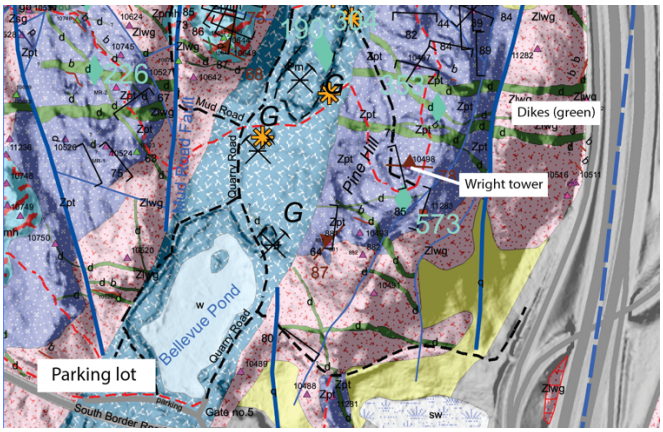
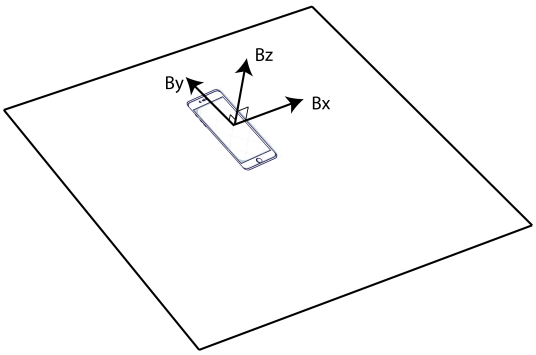


Figure 3: Geological map of the SE corner of the Middlesex Fells, showing our field area and the dikes the magnetism of which we attempt to measure. Map credit Ridge (2023) The Bedrock Geology of the Middlesex Fells Reservation

**Detailed instructions and tips**

- **Preliminaries.** You'll use the Physics Toolbox Suite to make the actual measurements. Please become familiar with starting and stopping measurements, saving data, and moving your phone along the profile guide device before you get into the field. Figure out the x, y, z axes definitions for your phone magnetometer. My iPhone has this configuration:




- **Quantifying uncertainty.** This is an **especially important component** of this project! Think through with your group the different sources of error, how important they might be, and what you can do to mitigate and/or quantify them. Some sources might include: phone magnetometer noise, uneven speed of traverse, unsteady orientation of phone during traverse, uneven spacing or orientation of guide, uneven distance between the guide and the rock. There are more! Think of experiments you can run to quantify these. Such measurements are often called "blanks" in research papers.
- **Deciding on your mapping locations.** The teaching staff will guide you to an outcrop of dikes in the Middlesex Fells. Before starting the bulk of your measurements, use your phone hand-held and survey the area, noting the qualitative morphology of the magnetic anomalies and their locations. Note how these locations relate to the contact

between the dike and surrounding rock. You might want to use masking tape but be sure to remove all of it before you leave!

- **Measurements.** Teaching staff will be on hand to advise at the Fells. Basic format will likely involve regularly spaced magnetic field profiles over a defined region. Before/while you take this data, it's important to **take notes**. These should include: sketch of the measurement geometry including orientation, sketch of the geometry of the dike and any observations of dike-surrounding rock contacts, definitions of x, y, z, scanning direction, scan velocity, method of maintaining scan geometry and speed, correspondence to key rock features, photos of setup, and anything else you think of. Store your data and make sure everything is saved before you move on!
- **Analysis.** The core of this particular project is to use the magnetic field data you collected to discovery the magnetization direction of the dike and all its geological implications. This, in essence, is a data processing and non-linear regression problem. Whatever you do, your project should **demonstrate thoughtful treatment of uncertainty** and error propagation. Here are some suggested topics and associated analysis types:
    - Non-linear regression to quantify the magnetization direction, which will require simultaneous fits for some aspects of dike geometry that you cannot measure.
    - Using your multiple profiles to constrain the uncertainties related to measurement and local rock geometry.
    - Propagating your uncertainties from the magnetization direction to the latitude and ultimately age using analytic or Monte Carlo methods.

**Some other things to think about**

- How much do we know about the geometry of the dike?
- How "unique" is your best-fit solution? In other words, can you achieve a similar degree of fit using a different set of parameters and assumptions, for example about dike geometry?
- How does small scale topography on the rock surface affect your measurements? Can you use your multiple profiles to constrain this effect?

**Primer on mathematics of magnetic fields arising from a subsurface volume:**

The good news is that the magnetic field signal of dikes is well-studied as magnetic surveys from both the ground and aircraft routinely pick up these signatures.

The bad news is that, despite lots of efforts to simplify the mathematics, this is inherently a somewhat messy problem.

We will not go through any derivations here- suffice to say that for uniformly magnetized volumes that can be reduced to a set of planar surfaces, the magnetic field it generates can be computed using a series of integrals over all the surfaces. In one of the simplest possible configurations, the vertical component of the magnetic field (with respect to the measurement plane) generated by a vertical, infinitely long dike is (see Fig. 4 for geometry):

$$B_z(x) = 2J\left\{\cos\phi \log\left[\frac{(-W/2+x)^2+Z^2}{(W/2+x)^2+Z^2}\right] + \sin\phi\left[\tan^{-1}\frac{W/2+x}{Z} - \tan^{-1}\frac{-W/2+x}{Z}\right]\right\}$$

Where $x$ is the independent variable for distance along the profile and $J$ is a material property of the rock related to the concentration of magnetic minerals. In this equation, $x = 0$ is defined as the point on the profile directly above the center of the dike. Remember this $B_z$ refers to the field created by the dike- you'll have to subtract off the magnetic field of the Earth as a background value.

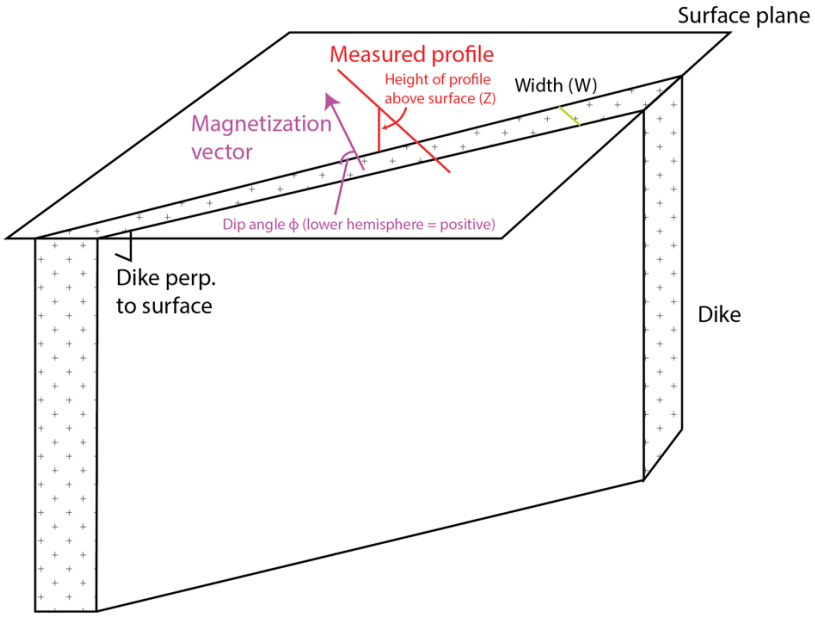
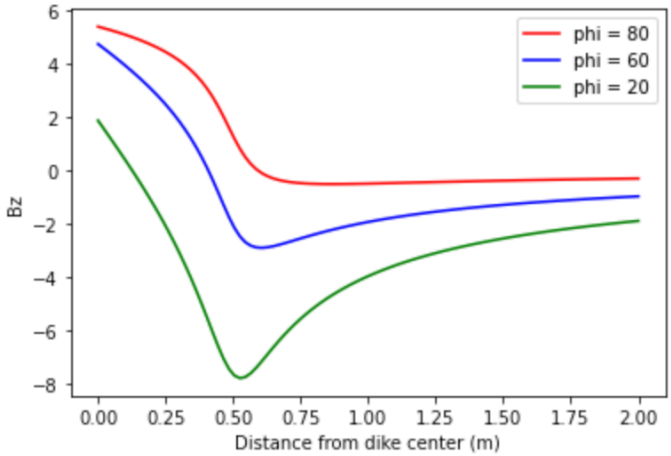



Figure 4: Example geometry of a dike and measurement profile.

Here are the predicted magnetic fields for a few different dips of the magnetization, which shows how the data can potentially distinguish among these. The other parameters used there are $W = 1$ m, $Z = 0.1$ m.




Note that this is really for the simplest geometry. If the dike is non-vertical or if the magnetization has a direction with a different azimuth relative to the measurement profile, the equation becomes more complicated. For a more complete treatment, please see, in order of complexity, Murthy (1990), Bhattacharyya (1964), Hjelt(1974). These are available on Canvas.

The equation plotted above is a simplified version of Eq. 4 in Murthy (1990). If this simple equation does not yield a satisfactory fit to the data, you may need to dig into these works for more complete equations!

Finally, note that, beside the exact angle (and therefore latitude) of the magnetization, whether the magnetization points up or down from the horizontal can tell us the reversal state of the Earth's magnetic field! See Fig. 1.